MAHATMA GANDHI UNIVERSITY

III SEMESTER- FYUGP- PRACTICAL EXAMINATION 2025

B.Sc BIOLOGICAL TECHNIQUES AND SPECIMEN PREPARATION

MG3DSCBTS200- ESSENTIALS IN BIOLOGICAL TECHNIQUES- DSC A

Time: 2hrs **Maximum marks: 35**

I. Perform Agarose Gel Electrophoresis of the given sample (8 marks) **[CO5] [A]**

a) Principle (2 marks),

b) procedure (2 marks),

c) gel loading (4 marks)

or

Perform Paper Chromatography of the given amino acids (8 marks) **[CO5] [A]**

a) Principle (2 marks),

b) procedure (2 marks),

c) sample spotting/ handling (4 marks)

II. Radiology Problems: (2x4=8 marks) **[CO6] [A]**

i) a) Half-life of Uranium^{235} is 7.04 x 10^{8} years. 12.5 gm of U^{235} remains after 2.82 x 10^{9} years. How much of the sample radioactive isotope was there in the original sample?

or

b) What is the half-life of 100 g of N^{16} that decays to 12.5 g in 21.6 sec?

ii) a) 800 g sample of technetium decays to 200g in 639000 years. What is the half-life?

or

b) half-life of the materials is 3 hours. If the material count rate is 544 becquerel (Bq). What will be the count rate after 15 hours?

III. Identify and write one point of importance of the following: (4 marks) [CO5] [U]

IV. Certified Record (10 marks)

V.Viva (5 marks) **[CO5]**

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